Corrections to `Calculation_main.tex`

Equations are quoted by their labels in `Calculation_main.tex`, with the line number in brackets. Red is the text as written; blue is the correction. The only longitudinal conditions used are $\vee > p^+, k^+ > \Lambda$ for every gluon: no limit $\Lambda \to 0$ or $\vee \to \infty$ is taken anywhere. Every statement is checked numerically or symbolically (Sec. 7). The diagrams are ignored. A copy of the file with all corrections marked in place (red/blue) is `Calculation_main_corrected.tex` (and `.pdf`).

How it was checked.

- Every wave function was rebuilt from the matrix elements Eq. (`m1`)–Eq. (`m5`) with Rayleigh–Schrödinger perturbation theory, and compared point by point with the notes before any integration. This uses SU(3) structure constants, random momenta and a factor $\frac{1}{2}$ per identical two-gluon state.
- The transverse loop integrals were redone in dimensional regularisation ($d = 2 - \epsilon$, $\overline{\text{MS}}$) with Feynman parameters.
- Every p^+ (or ξ) integral was redone exactly and compared with direct quadrature.
- The coefficients of Ω were compared with the verified wave functions.

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1 Summary

	where	what
Errors that change results		
E1	Eq. (`onegrho12`) [683]	overall sign: $-\frac{ig^2 f^{abc}}{8\pi^3} \to +\frac{ig^2 f^{abc}}{8\pi^3}$. Its ξ-form below is right.
E2	Eq. (`Eq:onegrhoX1`) [701]	overall sign: $+\frac{ig^2 f^{abc}}{4\pi^3} \to -\frac{ig^2 f^{abc}}{4\pi^3}$.
E3	$\mathbb{B}_3$, momentum and both coordinate forms [1276–1300]	the non-instantaneous $\Theta(p^+ - k^+ - \Lambda)$ entry needs a minus sign. Then the $1/\Lambda$ terms cancel, as the remark after Eq. (`Eq:onegrhoX2`) requires.
E4	$\mathbb{A}_2^{(3)}$, momentum space [1252]	the one-ρ part counts $\frac{\pi^2}{3} - \frac{67}{9}$ twice and misses the pole of $\lvert\Psi^B_{g\rho}\rangle$. It must be the bracket of Eq. (`grhoposition`).
E5	unitarity, $\mathbb{A}_1^{(3)\dagger}$ [1087–1101]	$\mathbb{N}^{(2)}\mathbb{A}^{(1)} = \mathbb{A}^{(1)}\mathbb{N}^{(2)}$ is false. Keep $-[\mathbb{N}^{(2)}, \mathbb{A}_2^{(1)}]$.
E6	$\mathbb{B}_2$ in the (k^+, p^+) forms [1260, 1268]	$\Theta(k^+ - \Lambda - p^+) \to \Theta(p^+ - \Lambda)\Theta(k^+ - \Lambda)\Theta(\vee - k^+ - p^+)$.
E7	Appendix F [1857–1922]	the matrix $(U^\dagger - U^\dagger(t))(U - U(t))$ must sit at the points of $\mathcal{K}(u, v, t)$; the overall sign of $dN^{(5)}$ is $+$.
Typos (the next line is right)		
E8	Eq. (`part2`) [270]	second m_2 term: $\delta^{ni} \to \delta^{ji}$.
E9	Eq. (`stseve`) [355]	energy denominators: $\frac{\mathbf{q}^2}{2q^+}\left(\frac{\mathbf{r}^2}{2r^+} + \frac{\mathbf{p}^2}{2p^+}\right) \to \frac{\mathbf{p}^2}{2p^+}\left(\frac{\mathbf{r}^2}{2r^+} + \frac{\mathbf{q}^2}{2q^+}\right)$.
E10	Eq. (`onegrho22`) [731]	$\frac{k^+ + p^+}{k^+ - p^+} + (k-p)^l \to \frac{k^+ + p^+}{k^+ - p^+}(k-p)^l$.
E11	Eq. (`A1coeff`) [1247]	$-\mathbb{A}^a_{2i}(k^+, \mathbf{z}) \to -\mathbb{A}^a_{2i}(k^+, -\mathbf{z})$.
Exact forms (replace every $\Lambda \to 0$, $\vee \to \infty$ step)		
X1	Eq. (`gtwoloop1`) [201]	exact ξ integral (Sec. 3.1).
X2	Eq. (`instgg`) [222], Eq. (`grho3`), Eq. (`tworho2`)	$\frac{k^+}{\Lambda} - 1 - \ln\frac{\Lambda}{k^+} \to \frac{k^+}{\Lambda} - \frac{k^+}{k^+ - \Lambda} + \ln\frac{k^+ - \Lambda}{\Lambda}$.
X3	Eq. (`onerho2`) [277]	exact ξ integral (Sec. 3.3).
X4	Eq. (`tworho2`) [280], Eq. (`combinedgrhorho`) [524]	exact; in the sum with Eq. (`grho3`) only $\ln\frac{\Lambda}{k^+} \to \ln\frac{\Lambda}{k^+ - \Lambda}$ changes.
X5	Eq. (`threerhoafter`) [411], after Eq. (`zetainteg`) [432]	no approximation is needed: the $\frac{1}{\Lambda} - \frac{1}{\vee}$ terms cancel exactly.
X6	Eq. (`combinedgrho`), Eq. (`grhoposition`) [496–518]	exact ξ limits $[\frac{k^+}{\vee}, \frac{k^+}{k^+ + \Lambda}]$, $k^+ < \vee - \Lambda$ for the Ψ^B part; exact bracket.
X7	after Eq. (`combinedgrhorho`) [533], Eq. (`combinedgrhorhopos`) [540]	exact ξ limits; keep Λ in $\ln\frac{\Lambda\mathbf{k}^2 + k^+\mathbf{p}^2}{\vee\mathbf{k}^2 + k^+\mathbf{p}^2}$.
Only for a general adjoint U (with your convention $U^{bc} = U^{cb}$ the two forms are equal)		
G1	combined $\mathcal{A}^{ab}_{ij}$, $\mathbb{X}_1$ [1490, 1494]	$U^{\dagger db}(v) \to U^{db}(v)$.
G2	Eq. (`eq: A1a`) and the first combined $\mathcal{A}^{ab}_{ij}$ [1468, 1490]	$U^{bc}(v) \to U^{\dagger bc}(v) = U^{cb}(v)$.
G3	item $\bar{\mathbb{C}}\mathbb{A}$ [1484]	$U^{bd}(v) \to U^{\dagger bd}(v) = U^{db}(v)$.
G4	Eq. (`eq:CS2`) [1538]	seven placements (Sec. 4).
Minor		
M1–M8		notation and typography (Sec. 5).

2 Errors that change results

2.1 E1, E2: sign of the non-instantaneous term with $p^+ > k^+$

Rebuilt from Eq. (`m2`), Eq. (`m3`) and $\langle p|H|qr\rangle = \langle qr|H|p\rangle^*$, Eq. (`onegrho12`) has the wrong overall sign, while its ξ-form (the next equation) is right ($-\frac{ig^2 f^{abc}}{4\pi^3}$). Eq. (`onegrho22`) and its ξ-form are right.

$$|(\Psi^a_i(k))^1_{g\rho}\rangle = -+\frac{ig^2 f^{abc}}{8\pi^3}\int_{k^+ + \Lambda}^{\vee} dp^+ \int d^2p\, \frac{\rho^c(k-p)(p-k)^l}{\sqrt{p^+(p^+ - k^+)k^+}\,(p^+ - k^+)^{3/2}} \times \frac{1}{\left(\frac{\mathbf{k}^2}{k^+} - \frac{\mathbf{p}^2}{p^+}\right)\frac{(\mathbf{k}-\mathbf{p})^2}{k^+ - p^+}}\Big[\cdots\Big]\,|g^b_j(p)\rangle\,. \tag{1}$$

The Fourier transform of the (correct) ξ-form with Eq. (`i4`), at $X' = x - z'$, $X = x - z$, gives

$$|(\Psi^a_i(k^+, \mathbf{z}'))^1_{g\rho}\rangle = +-\frac{ig^2 f^{abc}}{4\pi^3}\int_{1+\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}} d\xi \int_{x,z} \frac{\sqrt{\xi}}{1-\xi}\,\frac{X'^m - \xi X^m}{X'^2 - \xi X^2}$$

$$\times \frac{X'^l - X^l}{(X'-X)^2}\,\rho^c(\mathbf{x})$$
$$\times \left(\delta_{il}\delta_{jm} - \frac{1}{\xi}\delta_{jl}\delta_{im} - \frac{1}{1-\xi}\delta_{ij}\delta_{lm}\right)|g^b_j(\xi k^+,\mathbf{z})\rangle\,. \tag{2}$$

Near $\xi = 1$ the $\delta_{ij}\delta_{lm}/(1-\xi)$ part of this term now has the sign opposite to the instantaneous term Eq. (`Eq:onegluoninstX1`), so the $(1-\xi)^{-2}$ endpoint singularity cancels for $\xi \to 1^+$ as it does for $\xi \to 1^-$. With the sign as written the two add, and a term $\propto k^+/\Lambda$ survives.

2.2 E3: $\mathbb{B}_3$

$\mathbb{B}_3$ was built from Eq. (`onegrho12`) and Eq. (`Eq:onegrhoX1`), so the $\Theta(p^+ - k^+ - \Lambda)$ entry of the non-instantaneous term inherits E1/E2. Compared with the first-principles wave functions, the two non-instantaneous entries as written have ratios $-8\pi^3$ ($p^+ > k^+$) and $+8\pi^3$ ($p^+ < k^+$); $8\pi^3$ is the same factor as for $\mathbb{B}_2$. Corrected:

$$\text{momentum:}\quad \Big[+-\frac{k^+-p^+}{(\mathbf{k}-\mathbf{p})^2}\,\Theta(p^+-k^+-\Lambda) + \frac{\Theta(k^+-\Lambda-p^+)}{\frac{\mathbf{k}^2}{k^+} - \frac{(\mathbf{k}-\mathbf{p})^2}{k^+-p^+} - \frac{\mathbf{p}^2}{p^+}}\Big], \tag{3}$$

$$\text{coordinate, } (k^+,p^+):\quad \Big[+-\frac{\Theta(p^+-k^+-\Lambda)}{((x-z)-(x-u))^2} - \frac{p^+k^+\,\Theta(k^+-\Lambda-p^+)}{(k^+(x-z)-p^+(x-u))^2}\Big], \tag{4}$$

$$\text{coordinate, } \xi:\quad \Big[+-\frac{\Theta(\xi-1-\frac{\Lambda}{k^+})}{((x-z)-(x-u))^2} - \frac{\xi\,\Theta(1-\frac{\Lambda}{k^+}-\xi)}{((x-z)-\xi(x-u))^2}\Big]. \tag{5}$$

Everything else in $\mathbb{B}_3$ is unchanged. With the sign as written, Group I Row III of the cross section has a term $\propto 1/\Lambda$; with the corrected sign every $1/\Lambda$ term cancels.

2.3 E4: $\mathbb{A}_2^{(3)}$ in momentum space

The one-ρ part of $\mathbb{A}_2^{(3)}(k^+,k)$ [1252] is written as the bracket of $|\Psi^A_{g\rho}\rangle$ plus a second bracket. That second bracket, $\frac{\pi^2}{3} - \frac{67}{9} + \ln^2(1-\frac{k^+}{\vee}) + \ln(1-\frac{k^+}{\vee})\ln\frac{\vee}{k^+}$, is already the finite part of the total $|\Psi^A_{g\rho}\rangle + |\Psi^B_{g\rho}\rangle$ (it is what Eq. (`grhoposition`) uses). So $\frac{\pi^2}{3} - \frac{67}{9}$ is counted twice, $\ln^2\frac{\Lambda}{k^+} - \ln^2\frac{\vee}{k^+}$ is left over, and the pole of $|\Psi^B_{g\rho}\rangle$ is missing. The two lines must be replaced by the momentum-space version of Eq. (`grhoposition`):

$$\begin{aligned}
\mathbb{A}^{(3)a}_{2i}(k^+,k) \supset\ & \frac{g^3N_c\rho^a(-k)k^i}{8\sqrt{2k^+}\pi^2k^2}\Big[\Big[-\frac{2}{\epsilon} + \ln\frac{\mathbf{k}^2}{\mu^2_{\overline{MS}}}\Big]\Big[\frac{11}{3} + \ln\frac{\Lambda}{k^+} - 2\ln\frac{\vee}{\Lambda}\Big] + \frac{\pi^2}{3} - \frac{67}{9} + \ln^2\frac{\Lambda}{k^+} - \ln^2\frac{\vee}{k^+}\Big] \\
& + \frac{g^3N_c\rho^a(-k)k^i}{8\sqrt{2k^+}\pi^2k^2}\Big[\frac{\pi^2}{3} - \frac{67}{9} + \ln^2\Big(1-\frac{k^+}{\vee}\Big) + \ln\Big(1-\frac{k^+}{\vee}\Big)\ln\frac{\vee}{k^+}\Big] \\
=\ & \frac{g^3N_c\rho^a(-k)k^i}{8\sqrt{2k^+}\pi^2k^2}\Big[\Big[-\frac{2}{\epsilon} + \ln\frac{\mathbf{k}^2}{\mu^2_{\overline{MS}}}\Big]\Big[\frac{11}{3} + \ln\frac{\Lambda}{k^+} + \ln\Big(\frac{\vee}{k^+}-1\Big) - \ln\frac{\vee}{\Lambda}\Big] \\
& + \frac{\pi^2}{3} - \frac{67}{9} + \ln^2\Big(1-\frac{k^+}{\vee}\Big) + \ln\Big(1-\frac{k^+}{\vee}\Big)\ln\frac{\vee}{k^+}\Big].
\end{aligned} \tag{6}$$

This is the $\Lambda \to 0$ form, as in Eq. (`grhoposition`); the exact form is in Sec. 3.6. The coordinate-space $\mathbb{A}_2^{(3)}$ [1256] already has the right bracket.

2.4 E5: unitarity and $[\mathbb{N}^{(2)}, \mathbb{A}^{(1)}]$

The step "$\mathbb{A}^{(1)}$ is linear in ρ, hence $\mathbb{N}^{(2)}\mathbb{A}^{(1)} = \mathbb{A}^{(1)}\mathbb{N}^{(2)}$" [1092–1096] is not right. The coefficient

$$\mathbb{N}^{(2)} = -\frac{g^2}{(2\pi)^3}\ln\frac{\vee}{\Lambda}\int_{x,y} K(x,y)\rho^a(x)\rho^a(y), \qquad K(x,y) = \int_z \frac{(x-z)\cdot(y-z)}{(x-z)^2(y-z)^2} = K(y,x), \tag{7}$$

commutes with the total charge, but not with $\rho^b(u)$:

$$\Big[\int_{x,y} K(x,y)\rho^a(x)\rho^a(y),\ \rho^b(u)\Big] = if^{abc}\int_x K(x,u)\,\{\rho^a(x),\rho^c(u)\} \neq 0. \tag{8}$$

So the relation must be kept as derived just above [1087]:

$$\mathbb{A}^{(3)\dagger b}_{1j}(p) = -\mathbb{A}^{(3)b}_{2j}(p) - [\mathbb{N}^{(2)}, \mathbb{A}^{(1)b}_{2j}(p)]$$

$$+\int\frac{d^3q}{(2\pi)^3}\Big[2\mathbb{A}^{(1)c}_{1k}(q)\mathbb{B}^{cb}_{2kj}(p,q)-\mathbb{B}^{\dagger cb}_{3kj}(p,q)\mathbb{A}^{(1)c}_{2k}(q)-2\mathbb{C}^{\dagger dcb}_{1lkj}(p,q)\mathbb{B}^{cd}_{2kl}(q,p-q)\Big], \tag{9}$$

and the "simplified" relation [1099–1101] must be dropped. The commutator is a two-charge term $\propto \ln\frac{\vee}{\Lambda}$. It enters only the $\ln\frac{\vee}{\Lambda}$ bookkeeping of Group I: it is the α part of the $U(w)\mathbb{A}^{(3)}$ terms of Row II (fix F4 of `ThreeRho_check.pdf`).

2.5 E6: range of $\mathbb{B}_2$

In $\mathbb{B}^{ba}_{2ji}(k^+,k;p^+,p)$ [1260] and in the coordinate form $\mathbb{B}^{bc}_{2ji}(k^+,-w;p^+,-z)$ [1268], k^+ and p^+ are the momenta of the two gluons. $\Theta(k^+-\Lambda-p^+)$ is the ξ-form $\Theta(1-\frac{\Lambda}{k^+}-\xi)$ (where k^+ is the total momentum) carried over unchanged:

$$\Theta(k^+-\Lambda-p^+)\;\to\;\Theta(p^+-\Lambda)\,\Theta(k^+-\Lambda)\,\Theta(\vee-k^+-p^+) \tag{10}$$

in all six places. Your Group II and III notes already use $\Lambda<p^+<\vee-k^+$. The momentum-space $\mathbb{B}_2$ is otherwise right: it equals $8\pi^3$ times Eq. (`psiggrhomom`)+Eq. (`ggrhorhomom`) term by term.

2.6 E7: Appendix F

These are the two points reported before (`AppEF_check.pdf`); Appendix E is right.

- In $dN^{(1)}$–$dN^{(4)}$ every line carries the same matrix $(U^\dagger(x)-U^\dagger(t))(U(y)-U(t))$ or its transpose. It must sit at the points u,v of the kernel $\mathcal{K}(u,v,t)$ of that line, where the two J_R act:
  - $dN^{(1)}$, $\mathcal{K}(u,v,t)$: $(U^\dagger(u)-U^\dagger(t))(U(v)-U(t))$, i.e. (y,x), (w',x), (y,w), (w',w) in the four lines;
  - $dN^{(2)}$, $\mathcal{K}(u,v,t)$: the transpose $(U^\dagger(v)-U^\dagger(t))(U(u)-U(t))$, i.e. (y,x) [as printed], (w',x), (y,w), (w',w);
  - $dN^{(3)}$, $dN^{(4)}$, $\mathcal{K}(v,v,t)$: $(U^\dagger(v)-U^\dagger(t))(U(v)-U(t))$ with $v=y,w'$ and $v=x,w$.
- $dN^{(5)}$: overall sign $-+\frac{\alpha_s^2\delta Y}{8\pi^6(N_c^2-1)}$; Eq. (`Eq:HLOfinal`) itself is right. Also the second factor of each line is a trace: $\{\dots\}\to\mathrm{Tr}\{\dots\}$.

2.7 E8–E11: typos

- E8, Eq. (`part2`): in the second m_2 term the gluon p is emitted first, so $\delta^{ab}\delta^{ni}\delta^{ji}\delta^{(3)}(q-p)\frac{g(\mathbf{k}^n-\mathbf{p}^n)\rho^c(-\mathbf{k}+\mathbf{p})}{4\pi^{3/2}|k^+-p^+|^{3/2}}$. As written, n occurs three times. Eq. (`3grhorho`) is right.
- E9, Eq. (`stseve`): $\frac{1}{(\frac{\mathbf{q}^2}{2q^+})(\frac{\mathbf{r}^2}{2r^+}+\frac{\mathbf{p}^2}{2p^+})(\frac{\mathbf{p}^2}{2p^+})(\frac{\mathbf{r}^2}{2r^+}+\frac{\mathbf{q}^2}{2q^+})(\frac{\mathbf{k}^2}{2k^+})}$, i.e. $E_g(p)E_{gg}(r,q)E_g(k)$ as in Eq. (`3rho`). The next line is right.
- E10, Eq. (`onegrho22`): $\Big(\frac{k^++p^+}{k^+-p^+}+(k-p)^l-p^l-k^l\Big)\to\Big(\frac{k^++p^+}{k^+-p^+}(k-p)^l-p^l-k^l\Big)$.
- E11, Eq. (`A1coeff`): $-\mathbb{A}^a_{2i}(k^+,\mathbf{z}-\mathbf{z})=\mathbb{A}^{\dagger a}_{1i}(k^+,\mathbf{z})$, as in the unitarity relation $\mathbb{A}^{(1)\dagger}_1(p^+,z)=-\mathbb{A}^{(1)}_2(p^+,-z)$. The formulas used afterwards are consistent with this.

3 Exact longitudinal integrals

Below, $\delta\equiv\Lambda/k^+$, $P_\epsilon\equiv-\frac{2}{\epsilon}+\ln\frac{\mathbf{k}^2}{\mu^2_{\overline{MS}}}$, and every bracket multiplies the prefactor written in the notes. In the coordinate forms $P_\epsilon\to-\frac{2}{\epsilon}-2\gamma_E-\ln\frac{(x-z)^2\mu^2}{4}$ by Eq. (`i1`), Eq. (`i8`), as in the notes. Each closed form below agrees with direct quadrature to 13–15 digits (Sec. 7), and its $\delta\to0$ limit is the notes' expression.

3.1 X1: Eq. (`gtwoloop1`)

With $w(\xi)=-2+\xi(1-\xi)+\frac{1}{\xi(1-\xi)}$, the line before Eq. (`gtwoloop1`) is $-2\int_\delta^{1-\delta}d\xi\,w(\xi)[P_\epsilon+\ln\xi(1-\xi)]$ in units of $\frac{g^3N_c\rho^a(-\mathbf{k})\mathbf{k}^i}{32\pi^{7/2}\mathbf{k}^2}$. Exactly:

$$\Big(\frac{11}{3}+4\ln\delta\Big)P_\epsilon+2\ln^2\delta-\frac{67}{9}+\frac{2\pi^2}{3}$$
$$\to\Big(\frac{11}{3}-8\delta+2\delta^2-\frac{4}{3}\delta^3+4\ln\frac{\delta}{1-\delta}\Big)P_\epsilon-\frac{67}{9}+\frac{2\pi^2}{3}+\frac{46}{3}\delta-\frac{4}{3}\delta^2+\frac{8}{9}\delta^3$$

$$+\Big(\frac{22}{3}-8\delta+2\delta^2-\frac{4}{3}\delta^3\Big)\ln(1-\delta)-\Big(8\delta-2\delta^2+\frac{4}{3}\delta^3\Big)\ln\delta$$
$$-2\ln^2(1-\delta)+2\ln^2\delta-4\ln\delta\ln(1-\delta)-8\,\mathrm{Li}_2(\delta). \tag{11}$$

3.2 X2: Eq. (instgg), Eq. (grho3)

$\frac{k^+}{2}\int_\Lambda^{k^+-\Lambda} dp^+ \frac{p^++k^+}{p^+(k^+-p^+)^2}$ gives

$$\frac{k^+}{\Lambda}-1-\ln\frac{\Lambda}{k^+} \;\to\; E\equiv\frac{k^+}{\Lambda}-\frac{k^+}{k^+-\Lambda}+\ln\frac{k^+-\Lambda}{\Lambda} \tag{12}$$

in the integrated form of Eq. (instgg), in Eq. (grho3), in its (vanishing) one-ρ part, and in Eq. (tworho2) (where it enters as $2E(\mathbf{k}-\mathbf{p})^2\mathbf{p}^i$).

3.3 X3: Eq. (onerho2)

The commutator part of Eq. (3grhorho), integrated over $\mathbf{p}$ in dimensional regularisation, gives the ξ-integrand (units $\frac{g^3N_c\rho^a(-\mathbf{k})\mathbf{k}^i}{32\pi^{7/2}\mathbf{k}^2}$)

$$\frac{(2-\xi)\ln\xi+(1+\xi)\ln(1-\xi)+(1+\xi)P_\epsilon}{\xi(1-\xi)}. \tag{13}$$

Integrated over $\delta<\xi<1-\delta$:

$$|\Psi^3_{g[\rho,\rho]}\rangle=-\int_\Lambda^\vee\frac{dk^+}{\sqrt{k^+}}\int d^2\mathbf{k}\,\frac{g^3N_c\rho^a(-\mathbf{k})\mathbf{k}^i}{32\pi^{7/2}\mathbf{k}^2}\Big(3\ln\delta\,P_\epsilon+\frac{\pi^2}{3}+2\ln^2\delta$$
$$\to 3\ln\frac{\delta}{1-\delta}\,P_\epsilon+\frac{\pi^2}{3}+2\ln^2\delta-2\ln^2(1-\delta)-2\ln\delta\ln(1-\delta)-4\,\mathrm{Li}_2(\delta)\Big)\,|g_i^a(k^+,\mathbf{k})\rangle\,. \tag{14}$$

3.4 X4: Eq. (tworho2) and its sum with Eq. (grho3)

The anticommutator part of Eq. (3grhorho), integrated over $\delta<\xi<1-\delta$, is exactly

$$|\Psi^3_{g\{\rho,\rho\}}\rangle=\int_\Lambda^\vee\frac{dk^+}{\sqrt{k^+}}\int d^2\mathbf{k}\,d^2\mathbf{p}\,\frac{ig^3f^{abc}\{\rho^b(-\mathbf{p}),\rho^c(-\mathbf{k}+\mathbf{p})\}}{32\pi^{9/2}(\mathbf{k}-\mathbf{p})^2\mathbf{k}^2}\,V^i\;|g_i^a(k^+,\mathbf{k})\rangle\,, \tag{15}$$

$$V^i=(\mathbf{k}-\mathbf{p})^i\Big[\frac{2\,\mathbf{k}\cdot(\mathbf{k}-\mathbf{p})}{\mathbf{p}^2}(\ell-\mathcal{L})-2\Big(\frac{1}{\delta}-\frac{1}{1-\delta}\Big)\Big]$$
$$-\frac{2\,\mathbf{p}\cdot(\mathbf{k}-\mathbf{p})}{\Delta}\mathcal{L}\,\mathbf{k}^i+2(\ell+\mathcal{L})\,\mathbf{p}^i, \tag{16}$$

with $\ell=\ln\frac{k^+-\Lambda}{\Lambda}$, $\Delta=(\mathbf{k}-\mathbf{p})^2-\mathbf{p}^2$, $\mathcal{L}=\ln\frac{k^+(\mathbf{k}-\mathbf{p})^2-\Lambda\Delta}{k^+\mathbf{p}^2+\Lambda\Delta}$. Under $\mathbf{p}\to\mathbf{k}-\mathbf{p}$ the colour factor $f^{abc}\{\rho^b(-\mathbf{p}),\rho^c(-\mathbf{k}+\mathbf{p})\}$ is odd, so only the odd part of the kernel contributes. For the sum of Eq. (grho3) (with E of X2) and $|\Psi^3_{g\{\rho,\rho\}}\rangle$, the odd part is exactly

$$\frac{\ln\frac{\Lambda}{k^+-\Lambda}}{\mathbf{p}^2(\mathbf{k}-\mathbf{p})^2}\Big[\mathbf{k}^2\mathbf{p}^i-(\mathbf{k}\cdot\mathbf{p})\,\mathbf{k}^i\Big]\quad\text{(odd part)}, \tag{17}$$

in the normalisation $\frac{ig^3f^{abc}\{\dots\}}{32\pi^{9/2}\mathbf{k}^2}$. All $\frac{1}{\Lambda}$ terms and all $\mathcal{L}$ terms cancel identically. So the first term of Eq. (combinedgrhorho) and of the lines after it is exact once

$$\ln\frac{\Lambda}{k^+}\Big[\frac{\mathbf{k}^2\mathbf{p}^i}{(\mathbf{k}-\mathbf{p})^2}-\frac{(\mathbf{k}\cdot\mathbf{p})\mathbf{k}^i}{(\mathbf{k}-\mathbf{p})^2}\Big]\;\to\;\ln\frac{\Lambda}{k^+-\Lambda}\Big[\frac{\mathbf{k}^2\mathbf{p}^i}{(\mathbf{k}-\mathbf{p})^2}-\frac{(\mathbf{k}\cdot\mathbf{p})\mathbf{k}^i}{(\mathbf{k}-\mathbf{p})^2}\Big]. \tag{18}$$

Eq. (tworho2) as written agrees with the exact result up to terms that vanish linearly in δ.

3.5 X5: the three-ρ terms

Eq. (tworhoafter) is not an approximation: it is the line before it multiplied out (the text "setting $\Lambda=0$" can go). For the $\rho\rho\rho$ terms, keep the exact p^+ and ξ integrals:

$$|\Psi^{6c}_{g\rho\rho\rho}\rangle\propto-\Big(\frac{1}{\Lambda}-\frac{1}{\vee}\Big),\qquad|\Psi^{7}_{g\rho\rho\rho}\rangle\supset+\frac{2k^+}{\mathbf{k}^2}\Big(\frac{1}{\Lambda}-\frac{1}{\vee}\Big). \tag{19}$$

With the same prefactor $\frac{g^3\mathbf{k}^i\rho^b(\mathbf{p})\rho^b(-\mathbf{p})\rho^a(-\mathbf{k})\sqrt{k^+}}{8\pi^{9/2}\mathbf{k}^4}$ and the same order of the charges, they cancel exactly. So Eq. (threerhoafter) and the "$\approx$" after Eq. (zetainteg) are not needed, and Eq. (3rhosmom) is exact as it stands.

3.6 X6: one-ρ total, Eq. (combinedgrho) $\to$ Eq. (grhoposition)

$|\Psi^B_{g\rho}\rangle$ is written with the momentum k^+ of the parent; its outgoing gluon has ξk^+. In terms of the outgoing momentum (again called k^+):

$$\int_\Lambda^\vee \frac{dK^+}{\sqrt{K^+}} \int_{\frac{\Lambda}{K^+}}^{1-\frac{\Lambda}{K^+}} d\xi\, f(\xi)\,|g(\xi K^+)\rangle = \int_\Lambda^{\vee-\Lambda} \frac{dk^+}{\sqrt{k^+}} \int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^++\Lambda}} \frac{d\xi}{\sqrt{\xi}} f(\xi)\,|g(k^+)\rangle\,. \tag{20}$$

The notes use the upper limit 1 (in $\ln\xi$) or $1-\frac{\Lambda}{k^+}$ (in $\ln(1-\xi)$), and $\vee$ for k^+. Exactly (units $\frac{g^3 N_c \rho^a(-\mathbf{k})\mathbf{k}^i}{32\pi^{7/2}\mathbf{k}^2}$, $\Lambda < k^+ < \vee - \Lambda$):

$$\begin{aligned}
&P_\epsilon\Big[\frac{11}{3} + \ln\frac{\Lambda}{k^+} + \ln\Big(\frac{\vee}{k^+}-1\Big) - \ln\frac{\vee}{\Lambda}\Big] \\
&+\frac{\pi^2}{3} - \frac{67}{9} + \ln^2\Big(1-\frac{k^+}{\vee}\Big) + \ln\Big(1-\frac{k^+}{\vee}\Big)\ln\frac{\vee}{k^+} \\
\to\ &P_\epsilon\Big[\frac{11}{3} + \ln\frac{\Lambda^2}{k^{+2}-\Lambda^2} + \ln\Big(1-\frac{k^+}{\vee}\Big) - 8\delta + 2\delta^2 - \frac{4}{3}\delta^3\Big] \\
&+\frac{\pi^2}{3} - \frac{67}{9} + \ln^2\Big(1-\frac{k^+}{\vee}\Big) + \ln\Big(1-\frac{k^+}{\vee}\Big)\ln\frac{\vee}{k^+} + R_\delta,
\end{aligned} \tag{21}$$

$$\begin{aligned}
R_\delta = {}& \frac{46}{3}\delta - \frac{4}{3}\delta^2 + \frac{8}{9}\delta^3 + \Big(\frac{22}{3} - 8\delta + 2\delta^2 - \frac{4}{3}\delta^3\Big)\ln(1-\delta) - \Big(8\delta - 2\delta^2 + \frac{4}{3}\delta^3\Big)\ln\delta \\
&-2\ln\delta\ln(1-\delta) - 4\,\mathrm{Li}_2(\delta) - \ln^2(1+\delta) + \ln(1+\delta)\ln\delta.
\end{aligned} \tag{22}$$

Note that $\frac{11}{3} + \ln\frac{\Lambda}{k^+} + \ln(\frac{\vee}{k^+}-1) - \ln\frac{\vee}{\Lambda} = \frac{11}{3} + 2\ln\frac{\Lambda}{k^+} + \ln(1-\frac{k^+}{\vee})$, so the $\delta\to 0$ limit is the notes' bracket. For $\vee - \Lambda < k^+ < \vee$ only $|\Psi^A_{g\rho}\rangle$ contributes:

$$P_\epsilon\Big[\frac{11}{3} - 8\delta + 2\delta^2 - \frac{4}{3}\delta^3 + \ln\frac{\delta}{1-\delta} - 2\ln\frac{\vee}{\Lambda}\Big] + \frac{\pi^2}{3} - \frac{67}{9} + \ln^2\delta - \ln^2\frac{\vee}{k^+} + R_\delta + \ln^2(1+\delta) - \ln(1+\delta)\ln\delta. \tag{23}$$

The same brackets go into $\mathbb{A}_2^{(3)}$ (E4) in both spaces.

3.7 X7: two-ρ total, after Eq. (combinedgrhorho)

The ξ integral of the last line of Eq. (combinedgrhorho) uses the same change of variables as X6. With $\alpha^i = \frac{\mathbf{p}^2\mathbf{k}^i + \mathbf{k}^2\mathbf{p}^i}{(\mathbf{k}-\mathbf{p})^2}$ and $\beta^i = \frac{(2\mathbf{p}^2 - 2\mathbf{k}\cdot\mathbf{p} + \mathbf{k}^2)\mathbf{k}^i + \mathbf{k}^2\mathbf{p}^i}{(\mathbf{k}-\mathbf{p})^2}$, it is exactly (for $\Lambda < k^+ < \vee - \Lambda$; zero above)

$$-\alpha^i \ln\frac{\vee - k^+}{\Lambda} - \mathbf{k}^i \ln\frac{\vee}{k^+k^+ + \Lambda} + \beta^i \ln\Big[1 + \Big(\frac{\vee}{k^+}-1\Big)\frac{\mathbf{k}^2}{(\mathbf{k}-\mathbf{p})^2}\Big] - \beta^i \ln\Big[1 + \frac{\Lambda}{k^+}\frac{\mathbf{k}^2}{(\mathbf{k}-\mathbf{p})^2}\Big]. \tag{24}$$

Together with X4, the line after Eq. (combinedgrhorho) [533] and its Fourier transform [537] are exact. In Eq. (combinedgrhorhopos) [540] keep $\ln\frac{\Lambda\mathbf{k}^2 + k^+\mathbf{p}^2}{\vee\mathbf{k}^2 + k^+\mathbf{p}^2}$; do not replace it by $\ln\frac{k^+\mathbf{p}^2}{\vee\mathbf{k}^2 + k^+\mathbf{p}^2}$. The same holds in both forms of $\mathbb{A}_2^{(3)}$.

4 Only for a general adjoint Wilson line

A real adjoint Wilson line is orthogonal, $U^\dagger = U^T$, but not symmetric. With your convention $U^{bc} = U^{cb}$ the written and the corrected forms below are equal. They differ for a general U, and only the corrected forms agree with the first-principles evaluation of the cross section (Groups_II_III_check.pdf, M4 and M6). The rule: the unobserved gluon (b,j) is created after the target by $\bar a^{\dagger d} = U^{de} a^{\dagger e}$, which gives $U^{db}(v) = U^{\dagger bd}(v)$; the conjugate amplitude is the complex conjugate (U is real).

- G1, combined $\mathcal{A}^{ab}_{ij}$ (both forms), $\mathbb{X}_1$: $-U^{ac}(w')\bar{\mathbb{A}}^d_j(q^+,-v)U^{\dagger db}(v)U^{db}(v)\mathbb{A}^c_i(k^+,-w')$. The item $\bar{\mathbb{A}}\mathbb{A}$ [1480] already has $U^{db}(v)$.
- G2, Eq. (eq: A1a) and the first combined $\mathcal{A}^{ab}_{ij}$: $U^{bc}(v) \to U^{\dagger bc}(v)$. The second combined form already has $U^{\dagger bc}(v)$.
- G3, item $\bar{\mathbb{C}}\mathbb{A}$: $\bar{\mathbb{C}}^{dac}_{1jik}U^{bd}(v)U^{\dagger bd}(v)U^{ce}(z)\mathbb{A}^{\dagger e}_{1k}$, as in the combined forms.

- G4, Eq. (eq:CS2). With the amplitude pieces (measured gluon (a,i,k^+) at w, unobserved (b,j,p^+) at z)

$$\begin{aligned}
\mathfrak{B} &= U^{ac}(w)\mathbb{B}^{bc}_{2ji}(k^+,-w;p^+,-z) - U^{cb}(z)\bar{\mathbb{B}}^{ca}_{2ji}(k^+,-w;p^+,-z),\\
\mathbb{X}'_1 &= U^{cb}(z)\bar{\mathbb{A}}^c_j(p^+,-z)\bar{\mathbb{A}}^a_i(k^+,-w) - U^{ac}(w)\bar{\mathbb{A}}^d_j(p^+,-z)U^{db}(z)\mathbb{A}^c_i(k^+,-w),\\
\mathbb{X}'_2 &= U^{cb}(z)\bar{\mathbb{A}}^a_i(k^+,-w)\bar{\mathbb{A}}^c_j(p^+,-z) - \bar{\mathbb{A}}^a_i(k^+,-w)\mathbb{A}^b_j(p^+,-z),\\
\mathfrak{C}' &= \int_y \Big[\bar{\mathbb{C}}^{dac}_{1jik}(p^+ + k^+, y-w; p^+, w-z)U^{db}(z)U^{ce}(y)\mathbb{A}^e_k(p^+ + k^+,-y)\\
&\qquad - U^{db}(z)\bar{\mathbb{C}}^{dac}_{1jik}(p^+ + k^+, y-w; p^+, w-z)\bar{\mathbb{A}}^c_k(p^+ + k^+,-y)\Big],
\end{aligned} \tag{25}$$

  and their complex conjugates at $w \to w'$, the placements of Eq. (eq:CS2) that differ are

  - $\mathfrak{B}$, conjugate: $\mathbb{B}^{\dagger bc'}_{2ji}U^{c'a}(w')U^{ac'}(w')$ and $\bar{\mathbb{B}}^{\dagger c'a}_{2ji}U^{bc'}(z)U^{c'b}(z)$;
  - $\mathbb{X}_1$, conjugate: $-\mathbb{A}^{\dagger c'}_i U^{bd'}(z)U^{d'b}(z)\bar{\mathbb{A}}^{\dagger d'}_j U^{c'a}(w')U^{ac'}(w')$;
  - $\mathbb{X}_1$, amplitude: $U^{bc}(z)U^{cb}(z)\bar{\mathbb{A}}^c_j\bar{\mathbb{A}}^a_i$;
  - $\mathbb{X}_2$, conjugate: $\bar{\mathbb{A}}^{\dagger c'}_j\bar{\mathbb{A}}^{\dagger a}_i U^{bc'}(z)U^{c'b}(z)$;
  - $\mathbb{X}_2$, amplitude: $U^{bc}(z)U^{cb}(z)\bar{\mathbb{A}}^a_i\bar{\mathbb{A}}^c_j$;
  - $\mathfrak{C}$, amplitude: $U^{bd}(z)U^{db}(z)$ in both terms.

  These apply in every place where the pieces appear, in Group II and in Group III. The conjugate $\mathfrak{C}$ is right as written.

5 Minor points

M1 Dimensional regularisation steps after Eq. (gtwoloop) [192] and before Eq. (onerhopart) [372]. The integral done in d dimensions is over $\tilde{\mathbf{p}}$ (or $\mathbf{p}$), not $\mathbf{k}$. Write $\int d^2\mathbf{k}$ and $\int d^2\tilde{\mathbf{p}} \to (2\pi)^d\mu^\epsilon\int\frac{d^d\tilde{\mathbf{p}}}{(2\pi)^d}$, i.e. the prefactor $\frac{1}{\sqrt{\pi}}$ with $\int\frac{d^d\mathbf{k}}{(2\pi)^d}$ is $\frac{1}{4\pi^{5/2}}$ with $\int d^2\mathbf{k}$. For $d=2-\epsilon$ the scale factor is μ^ϵ, not $\mu^{2\epsilon}$. The lines after them are right.

M2 LO wave function [163, 167, 443]: "$= \backslash - \; -$" has a stray \-.

M3 Coordinate $\mathbb{A}^{(3)}_2$ [1256]: $\ln\frac{V}{\Lambda} \to \ln\frac{\vee}{\Lambda}$.

M4 The integral after Eq. (i9) [1705] has no right-hand side (it is not used). Eq. (i9) is used only in commented-out lines.

M5 Oscillator form of $\mathsf{H}_{g\,gg}$ [1636]: ", $+h.c.$" inside the bracket; h.c. applies to the whole bracket.

M6 General normalisation [563]: $(1+x)^{-1/2} = 1 - \frac{x}{2} + \frac{3}{8}x^2$, so $-\frac{1}{8}+\frac{3}{8}\langle\Psi^{\rm LO}|\Psi^{\rm LO}\rangle^2$; the bracket opens with $\Big[$ and closes with $\Big)$. This is order g^4 and not used.

M7 Fourier transform after Eq. (combinedgrhorho) [537]: $\Big\{$ is closed by $\Big]$.

M8 The text before Eq. (tworhoafter) [398] says "Setting $\Lambda = 0$", but nothing is set to zero there (X5).

6 Verified as written

- Matrix elements Eq. (m1)–Eq. (m5), Fourier transforms Eq. (i1)–Eq. (i8) (earlier checks).
- Vacuum, two gluons: Eq. (ggrhorho1) and the next line; Eq. (decomposition); Eq. (part 1); Eq. (part 2); Eq. (gg_rhorho); Eq. (ggmid) and the next line; Eq. (contgg); Eq. (instgg3); Eq. (contgg3); Eq. (psiggrhomom); Eq. (psiggrhomomre); Eq. (ggrhopos) (with Eq. (i2), Eq. (i7)); Eq. (ggrhorhopos).

- Vacuum, one gluon: LO; Eq. (`gtwoloop`) up to the $\Lambda \to 0$ step (X1); Eq. (`instgg`) before the p^+ integral; Eq. (`grho3`); Eq. (`eq:gg-uu`), Eq. (`eq:ACrhorho`), Eq. (`grhorho4`) and its vanishing one-ρ part; Eq. (`3grhorho`); Eq. (`gfour`), Eq. (`inst4`), Eq. (`psif`); Eq. (`psiff`); Eq. (`4+5tworho`); Eq. (`4+5onerho`) (pole, $\ln\mu$ and finite parts at every ξ); Eq. (`4+5tworhosimpl`); the three-ρ chain Eq. (`3rho`) to Eq. (`onerhopart`), Eq. (`tworhoafter`), the $\rho\rho\rho$ terms, Eq. (`degen`), Eq. (`virtual`), Eq. (`zetainteg`); the sums $|\Psi^A_{g\rho}\rangle$, Eq. (`combinedgrho`), Eq. (`combinedgrhorho`); Eq. (`3rhosmom`), Eq. (`3rhospos`); Eq. (`normmom`), Eq. (`normpos`).
- One gluon in: the instantaneous terms and their transforms; the ξ-forms of Eq. (`onegrho12`) and Eq. (`onegrho22`); Eq. (`Eq:onegrhoX2`); Eq. (`m1grr`)–Eq. (`gtworhoX2`); Eq. (`rhoindep`), Eq. (`gindeppos`); $\mathbb{C}_1$.
- Ω: $\mathbb{N}$, Eq. (`A1coeffM`); $\mathbb{B}_2$ in momentum space ($8\pi^3\times$ wave function) and in coordinate space ($\frac{2\pi}{k^+}\times$ wave function), apart from E6; $\mathbb{B}_3$ apart from E3.
- Cross section: the Group I amplitude; the signs of the Group III terms of Eq. (`eq:CS2`); the LO normalisation; Appendix E.

7 Checks

The scripts are in `groups23/main/`. `python3 run_checks.py` runs all of them and writes `main_checks.json`; `python3 make_note.py` writes this table.

check	result
onegrho12 / first principles	-1 (spread 6.0e-15)
onegrho12 xi-form / first principles	1 (spread 1.0e-14)
onegrho22 (typo read as product) / first principles	1 (spread 3.5e-14)
onegrho22 xi-form / first principles	1 (spread 3.1e-14)
$\mathbb{B}_3$ Θ(p+-k+-Λ) entry / wave function	-248.0502134 (spread 2.8e-13)
$\mathbb{B}_3$ Θ(k+-Λ-p+) entry / wave function	248.0502134 (spread 3.1e-13)
instgg / first principles	1 (spread 4.4e-16)
eq:gg-uu / first principles	1 (spread 3.3e-16)
3grhorho / first principles	1 (spread 6.7e-16)
psif / first principles	1 (spread 3.3e-16)
psiff / first principles	1 (spread 4.4e-16)
4+5tworho / first principles	1 (spread 4.4e-16)
$\mathbb{B}_2$ (momentum) one-rho / wave function	248.0502134 (spread 2.0e-13)
$\mathbb{B}_2$ (momentum) two-rho / wave function	248.0502134 (spread 1.1e-13)
Ψ^3 commutator, ξ-integrand (dim. reg.) minus closed form of X3	3.8e-124 (three ξ values)
Eq. (`4+5onerho`) at $\xi = 0.2$: pole, $\ln\mu$, finite	-3.6, 1.8, 2.629215981 (notes -3.6, 1.8, 2.629215981)
Eq. (`4+5onerho`) at $\xi = 0.5$: pole, $\ln\mu$, finite	-3, 1.5, 2.364288669e-18 (notes -3, 1.5, 0)
Eq. (`4+5onerho`) at $\xi = 0.7$: pole, $\ln\mu$, finite	-2.6, 1.3, -1.58307634 (notes -2.6, 1.3, -1.58307634)
Eq. (`tworho2`) minus exact (odd part), $\delta = 0.01$	7.86e-02 (exact $\sim$ 4.67e+02)
Eq. (`tworho2`) minus exact (odd part), $\delta = 0.0001$	4.30e-04 (exact $\sim$ 2.19e+04)
Eq. (`tworho2`) minus exact (odd part), $\delta = 1e-06$	4.38e-06 (exact $\sim$ 2.22e+06)
gtwoloop1 pole coeff (-2 I0)	exact -7.52700362596, quadrature -7.52700362596 (rel. 1e-31); notes ($\Lambda \to 0$) -7.284946
gtwoloop1 finite (-2 I1)	exact 14.7853783669, quadrature 14.7853783669 (rel. 1e-31); notes ($\Lambda \to 0$) 14.127519
onerho2 pole coeff	exact 8.01302678827, quadrature 8.01302678827 (rel. 2e-31); notes ($\Lambda \to 0$) 8.2137095
onerho2 finite	exact -17.6437106963, quadrature -17.6437106963 (rel. 2e-31); notes ($\Lambda \to 0$) -18.282096
instgg factor E	exact 17.056371994, quadrature 17.056371994 (rel. 7e-31); notes ($\Lambda \to 0$) 17.192449
6c: int dp+/p+^2	exact 9.04743083004, quadrature 9.04743083004 (rel. 0e+00); notes ($\Lambda \to 0$) 9.0909091
7: (1/k+) int 2 dxi/xi^2 (cancels 6c x2)	exact 18.0948616601, quadrature 18.0948616601 (rel. 0e+00); notes ($\Lambda \to 0$) 18.181818
Psi^B pole coeff	exact 7.80814935656, quadrature 7.80814935656 (rel. 1e-31); notes ($\Lambda \to 0$) 7.870848

Psi^B finite	exact -1.08050478565, quadrature -1.08050478565 (rel. 1e-30); notes ($\Lambda \to 0$) -0.90491099
change of variables (test function)	exact 0.0305876763408, quadrature 0.0305876763408 (rel. 0e+00)
total pole coeff	exact -2.39136573937, quadrature -2.39136573937 (rel. 5e-31); notes ($\Lambda \to 0$) -1.8859268
total finite	exact -3.2280500732, quadrature -3.2280500732 (rel. 7e-31); notes ($\Lambda \to 0$) -4.3487003
tworho2 V exact vs quad [0]	exact 38.2608263705, quadrature 38.2608263705 (rel. 2e-16); max diff 7.1e-15
grho3+tworho2 antisym vs closed [0]	exact 0.0996485441678, quadrature 0.0996485441678 (rel. 1e-14); max diff 8.2e-15; notes with $\ln \frac{\Lambda}{k^+ - \Lambda}$: diff 6.9e-15
tworho2 V exact vs quad [1]	exact 65.2450378373, quadrature 65.2450378373 (rel. 0e+00); max diff 0.0e+00
grho3+tworho2 antisym vs closed [1]	exact 0.312130485356, quadrature 0.312130485356 (rel. 5e-15); max diff 6.1e-15; notes with $\ln \frac{\Lambda}{k^+ - \Lambda}$: diff 5.4e-15
R xi-integral exact vs quad [0]	exact 0.627405306989, quadrature 0.627405306989 (rel. 2e-16); notes ($\Lambda \to 0$) 0.65941739; max diff 3.3e-16
R xi-integral exact vs quad [1]	exact 0.659104337395, quadrature 0.659104337395 (rel. 1e-15); notes ($\Lambda \to 0$) 0.69910973; max diff 6.7e-16

- `lcpt1.py`: one gluon → one gluon, non-instantaneous, from Eq. (`m2`), Eq. (`m3`) (E1, E10).
- `b3_check.py`: non-instantaneous entries of $\mathbb{B}_3$ (E3).
- `lcpt_vac.py`: vacuum one-gluon terms at order g^3 from Eq. (`m1`)–Eq. (`m5`) (E8 and the verified list).
- `b2mom_check.py`: momentum-space $\mathbb{B}_2$ against the vacuum two-gluon wave functions.
- `onerho_dimreg.py`: commutator parts of Eq. (`3grhorho`) and Eq. (`psif`)+Eq. (`psiff`), transverse integral in dimensional regularisation (X3, Eq. (`4+5onerho`)).
- `tworho2_check.py`: Eq. (`tworho2`) against the exact ξ integral (X4).
- `exact_forms.py`: all exact forms of X1–X7 against quadrature, at $k^+ = 1.7$, $\Lambda = 0.11$, $\vee = 23$.